

Multiple Choice (10 marks)

1. The _____ is anything which your search engine cannot search.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Surface web
 - (d) Deep Web
2. A hash function guarantees the integrity of a message. It guarantees that the message has not be
 - (a) Replaced
 - (b) Overview
 - (c) Changed
 - (d) Violated
3. When the data must arrive at the receiver exactly as they were sent, its called
 - (a) Message Condentiality
 - (b) Message Integrity
 - (c) Message Splashing
 - (d) Message Sending
4. A packet filter firewall filters at the
 - (a) Application or transport
 - (b) Data link layer
 - (c) Physical Layer
 - (d) Network or transport layer
5. The Message condentiality or privacy means that the sender and the receiver expect
 - (a) Integrity
 - (b) Condentiality
 - (c) Authentication
 - (d) Nonrepudiation

True / False (10 marks)

Answer True or False

6. True or False: A buffer overflow occurs when the program actively rejects requests after detecting that a buffer is full.
7. True or False: WPA 2 is widely considered to be more secure than WPA 3 due to its longer history of use in enterprise networks.
8. True or False: Unauthorized network access is not classified as a transport layer vulnerability.
9. True or False: A buffer-overflow attack involves injecting extra data into memory to exploit vulnerabilities and execute unauthorized actions.
10. True or False: The Metasploit framework simplifies the exploitation of vulnerabilities through an intuitive, point-and-click interface.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the reasons why a compiler cannot determine the absolute address of a local variable, considering its stack allocation and implications for function calls in cybersecurity contexts.
12. Discuss the classic security properties in cybersecurity, identifying the one that does not fit within this framework. Analyze the implications of your choice on overall system security.

End of Paper